Samir Varma

732-401-6181 | sv773@scarletmail.rutgers.edu | linkedin.com/in/samir-varma-1a1166281/ | github.com/SamirVarma01

Education

Rutgers University, New Brunswick - Honors College GPA: 3.936/4.000

B.S. in Computer Engineering, B.S. in Mathematics *Aug. 2023 – May 2027*

- Relevant Coursework: Distributed Deep Learning, Software Engineering, Machine Learning, Computer Architecture

Experience

Software Engineering Intern May 2026 – August 2026

Dropbox *San Francisco, CA*

- Built an agentic Quarterly Business Review deck-generation service in Python/Flask, orchestrating LLM narrative steps with LangGraph over live customer data, with Redis-backed state and Bazel-managed service builds.
- Engineered a cloud data-grounding layer using Databricks SQL and SQLAlchemy with 11 parameterized query sources for trailing-12-month YoY analytics, cutting QBR drafting time by 95%.
- Designed a conversational editing pipeline with parallel data grounding, structured LLM outputs, schema-validated block mutations, optimistic concurrency control, and audit logging, achieving a 92% task success rate.

Data Science Intern June 2025 – August 2025

AT&T *Middletown, NJ*

- Deployed an intrusion detection service using Isolation Forest models built with Scikit-learn to score 3M+ network telemetry records with low-latency inference, raising anomaly precision to 96% through GridSearchCV tuning.
- Architected a scheduled Apache Airflow ETL pipeline orchestrating ingestion, validation, and feature preprocessing from Azure Databricks, enabling reliable daily model refreshes.

Software Engineering Intern January 2025 – April 2025

Coco Labs *San Francisco, CA*

- Developed AI inference services using Python, FastAPI, PyTorch, Docker, and Google Cloud Platform, orchestrating computer vision workflows to automate user-driven image editing via scalable backend services.
- Engineered distributed image processing pipelines using Python, OpenCV, Redis, and GCS to batch-process 5,000+ images, improving throughput and enabling reliable large-scale computer vision inference.

Machine Learning Research Assistant June 2024 – April 2025

Rutgers University *Piscataway, NJ*

- Implemented Actor–Critic reinforcement learning agents in PyTorch within the CARLA autonomous driving simulator, designing modular policy and value networks that improved average episode reward by 14%.
- Developed an evaluation framework that ran DQN and Actor–Critic experiments across parallel simulation runs, logging reward curves, convergence, and policy stability for reproducible comparison.

Projects

Distributed Task Queue & Durable LLM Agent Runtime | *Python, FastAPI, Redis Streams, MySQL, Docker*

- Built a distributed task queue on Redis Streams and MySQL with at-least-once delivery, heartbeat-based crash recovery, and fencing tokens; lost 0 of 10K tasks across 26 worker kills and cut p50 latency 39× via blocking reads.
- Designed a durable LLM agent runtime on the queue that checkpoints every LLM and tool call to resume mid-run after crashes, guaranteeing at-most-once side effects via intent records and idempotency keys (0 duplicates across 50 concurrent approval/cancel races) with token and cost budgets across Claude and OpenAI.

PokeChat | *Next.js, Tailwind CSS, FastAPI, OpenAI API, Chart.js, Beautiful Soup*

- Developed a full-stack competitive Pokemon team analysis website using FastAPI backend with AI-powered team evaluation and Next.js frontend, implementing web scraping for real-time strategic data from Pikalytics.com
- Built comprehensive team validation and meta trend visualization tools with Chart.js and the OpenAI API, enabling players to analyze team compositions, strengths and weaknesses, and competitive trends.

NextLevel | *Next.js, TypeScript, MongoDB, AWS S3, AWS Lambda*

- Built a game review platform on Next.js API routes and MongoDB, serving search and metadata for 400K+ games via the IGDB API with cached OAuth tokens, and REST endpoints for reviews, comments, and likes.
- Implemented credential auth with NextAuth (bcrypt-hashed passwords, JWT sessions) and an S3-triggered AWS Lambda that resizes uploads with Sharp into thumbnail, standard, and avatar variants in a separate processed bucket.

Skills

Languages: Python, Go, SQL, TypeScript, JavaScript, Java, C/C++

Tools: FastAPI, Flask, Django, Node.js, PostgreSQL, MySQL, Redis, Airflow, Databricks, AWS, GCP, Docker, Bazel, LangGraph, LangChain, OpenAI/Anthropic APIs, PyTorch, TensorFlow, Scikit-Learn